Arxivists
Research Recommender

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Table of Contents

Introduction

Project Goal

- Develop a **hybrid research paper recommender** system that helps users discover relevant articles beyond traditional keyword search
- **Recommend** papers for further reading & **citation** using multiple signals including article content, citation relationships, categories, and latent similarity patterns
- Combine content-based, collaborative & graph approaches (TF-IDF, SVD, NeuMF, PageRank, etc) to generate more **accurate** and **personalized** recommendations
- Enable cross-domain discovery by surfacing relevant work from related fields that users may not search for directly

Introduction Cont.

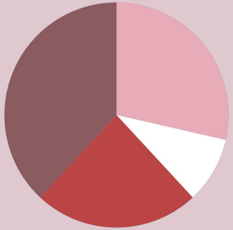
Business Value & Use case

- Reduces time spent manually searching through large academic databases
- Improves research quality by exposing users to broader and more relevant literature
- Supports interdisciplinary research by connecting ideas across domains
- Moves beyond exact keyword matching to recommend papers based on semantic similarity and citation behavior
- **Use case**: A user researching stock market modelling may receive recommendations not only from finance literature, but also machine learning papers on forecasting, time-series analysis, and related economic studies

Data Description

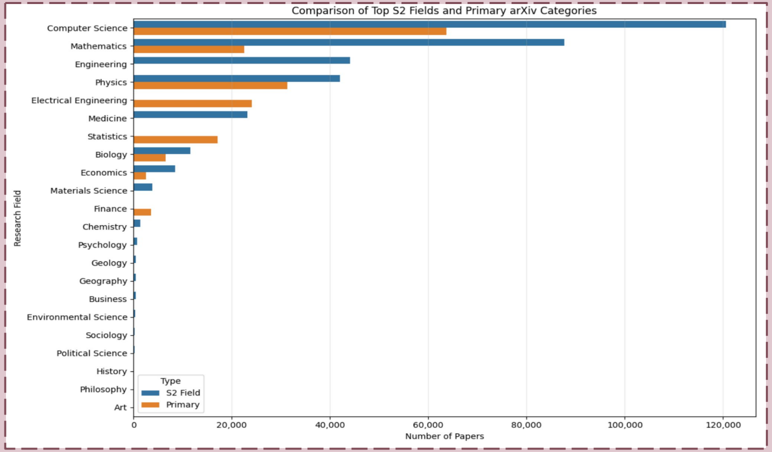
- Primary dataset: Cornell's arXiv dataset
- Contains approximately **1.7 million scholarly articles**
- Fields: Title, Abstract, Authors, Categories, Publication/ update date, etc
- Original data: **Physics (51%)**, followed by **Mathematics (22%)** & **Computer Science (19%)**, with smaller representation from other categories
- We chose **486,916** papers (1,000,638 citations/edges) from this original dataset and prioritised Computer Science, Physics & Math categories
- After applying the Cold start filter,
  - We have **171,289 papers, 951,597 citations**
  - A breakdown of: Computer Science (37.2%) becoming the dominant field, followed by Physics (18.3%), Electrical Engineering (14.1%), Mathematics (13.2%), and Statistics (10.0%), while Biology (3.8%), Finance (2.1%), and Economics (1.5%)

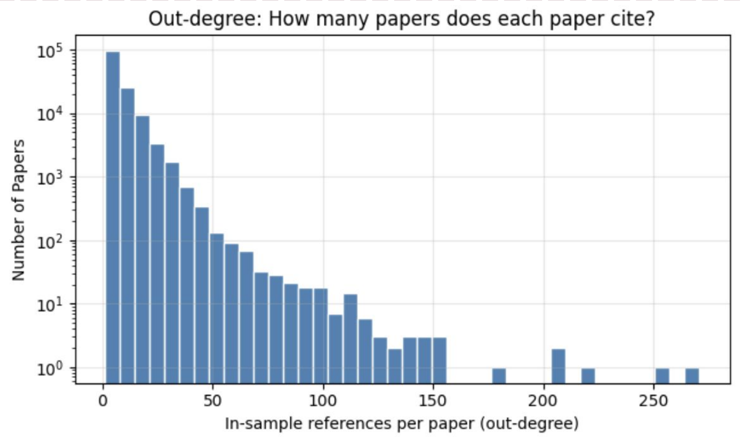
- The original arXiv dataset does **not include citation relationships**, which are required for collaborative and graph-based recommendation models
- Citation-enriched versions (unarXive) contain full text and citation structure but require additional access permissions
- Retrieved citation information through the **Semantic Scholar API**
- Constructed a separate **citation dataframe** containing: Source paper ID, Target (cited), and source of the cited paper paper ID using this library

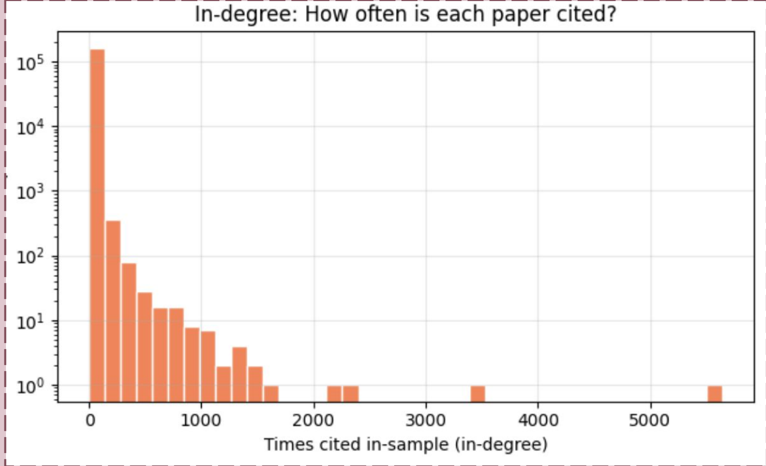


EDA *after cold start filter

- 171,289 papers, 951,597 citation links (edges)
- Articles Published Date: 2015-2025
- Median Abstract Length: 1120 words
- Citation Matrix (Sparsity) 99.9968%
- Cold start: 26.48% (papers never cited)
- Papers that cite others: 137,750 (80.4%)
- Papers that get cited: 123,989 (72.4%)
- On average, 6.91 out-degree, 7.67 in-degree citations







Methodology

Dataset Creation

- **Metadata collection:** Harvested arXiv metadata through **OAI-PMH** using parallelized collection & checkpointing for scalable and fault-tolerant ingestion
- **Citation enrichment:** Retrieved citation relationships from **Semantic Scholar API** in batched requests with rate limiting and incremental checkpoint recovery
- **Data filtering & cleaning:** Kept only citation edges where both papers existed in the collected dataset, applied ID standardization, and removed low-connectivity papers (degree < 2)
- **Dataset preparation:** Assigned integer paper indices for model compatibility and engineered features including weighted text representations, normalized citation popularity, and fields of study metadata
- **Train/test edge split:** Held out 20% of citation edges for evaluation while excluding training citations during recommendation (~800k vs 200k)

Feature Engineering

- **Text feature:** text = title + title + abstract (title duplicated to increase importance during text-based similarity)
- **Citation popularity:** cite_pop = normalized in-sample citation count (0–1)
- **Domain metadata:** s2_fields extracted from Semantic Scholar fields of study

Model Set-up

- **Citation graph construction:** Built a directed paper-paper graph where nodes represent papers & edges represent citation relationships (PageRank and Personalized PageRank)
- **Citation interaction matrix:** Converted citation edges into a sparse paper-paper interaction matrix for latent factor models (SVD, GMF & NeuMF)
- **Bandit reward setup:** Defined recommendation feedback using citation-based relevance, where successful recommendations correspond to recovering held-out citation links.

All Models

SVD (base)

Matrix factorization of the paper–paper citation matrix to learn latent citation relationships

TF-IDF

Computes cosine similarity using TF-IDF features extracted from paper title and abstract text

NeuMF

Learns nonlinear latent citation patterns using neural embeddings and interaction layers

PageRank

Computes global paper importance through recursive citation propagation across the citation graph

GMF

Learns non-linear paper embeddings using neural matrix factorization on citation interactions

Bandits

Balances exploitation of strong recommendations with exploration of potentially relevant unseen papers

Monte PPR

Estimates locally relevant papers through random walks starting from a query paper

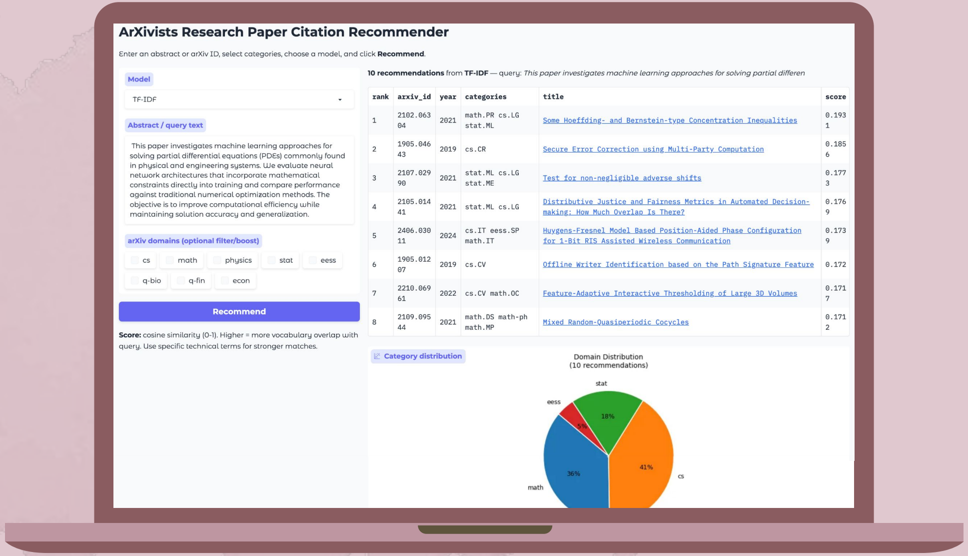
Hybrid 1

Weighted combination of SVD + TF-IDF

Hybrid 2

Weighted combination of TF-IDF + Monte Carlo PPR

Demo


ArXivists Research Paper Citation Recommender
Enter an abstract or arXiv ID, select categories, choose a model, and click Recommend.
Model
TF-IDF
Abstract / query text
This paper investigates machine learning approaches for solving partial differential equations (PDEs) commonly found in physical and engineering systems. We evaluate neural network architectures that incorporate mathematical constraints directly into training and compare performance against traditional numerical optimization methods. The objective is to improve computational efficiency while maintaining solution accuracy and generalization.
arXiv domains (optional filter/boost)
cs
math
physics
stat
eess
q-bio
q-fin
econ
Recommend
Score: cosine similarity (0-1). Higher = more vocabulary overlap with query. Use specific technical terms for stronger matches.
10 recommendations from TF-IDF — query: This paper investigates machine learning approaches for solving partial differen
rank | arxiv_id | year | categories | title | score
1 | 2102.06304 | 2021 | math.PR cs.LG stat.ML | Some Hoeffding- and Bernstein-type Concentration Inequalities | 0.1931
2 | 1905.04643 | 2019 | cs.CR | Secure Error Correction using Multi-Party Computation | 0.1856
3 | 2107.02990 | 2021 | stat.ML cs.LG stat.ME | Test for non-negligible adverse shifts | 0.1773
4 | 2105.01441 | 2021 | stat.ML cs.LG | Distributive Justice and Fairness Metrics in Automated Decision-making: How Much Overlap Is There? | 0.1769
5 | 2406.03011 | 2024 | cs.IT eess.SP math.IT | Huygens-Fresnel Model Based Position-Aided Phase Configuration for 1-Bit RIS Assisted Wireless Communication | 0.1739
6 | 1905.01207 | 2019 | cs.CV | Offline Writer Identification based on the Path Signature Feature | 0.172
7 | 2210.06961 | 2022 | cs.CV math.OC | Feature-Adaptive Interactive Thresholding of Large 3D Volumes | 0.1717
8 | 2109.09544 | 2021 | math.DS math-ph math.MP | Mixed Random-Quasiperiodic Cycles | 0.1712
Category distribution
Domain Distribution
(10 recommendations)
stat
eess
18%
41%
36%
math
cs

Applying models on a sample prompt

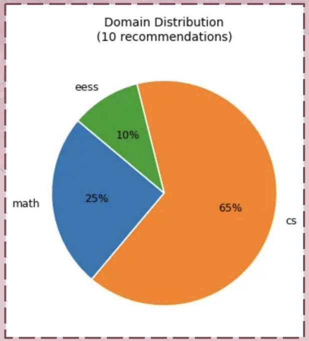
Given this prompt,

- abstract: This paper investigates machine learning approaches for solving partial differential equations (PDEs) commonly found in physical and engineering systems. We evaluate neural network architectures that incorporate mathematical constraints directly into training and compare performance against traditional numerical optimization methods. The objective is to improve computational efficiency while maintaining solution accuracy and generalization.
- Math + CS prompt
- No category boost added

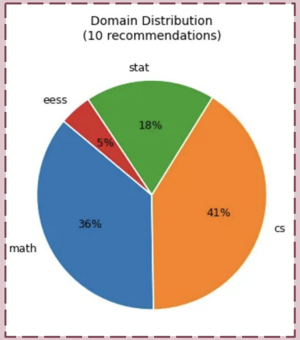
Top Recommendation from each model

1. SVD (Baseline): On an extension of the generalized BGW tau-function
2. TF-IDF: Some Hoeffding- and Bernstein-type Concentration Inequalities
3. NeuMF: Outside the Echo Chamber: Optimizing the Performative Risk
4. PageRank: U-Net: Convolutional Networks for Biomedical Image Segmentation
5. GMF: Lecture I: Governing the Algorithmic City
6. Bandits: SeiT++: Masked Token Modeling Improves Storage-efficient Training
7. Monte-PPR: Emergent family of Tsallis entropies from the q-deformed combinatorics
8. TF-IDF + SVD: Some Hoeffding- and Bernstein-type Concentration Inequalities
9. TF-IDF + MPPR: Secure Error Correction using Multi-Party Computation

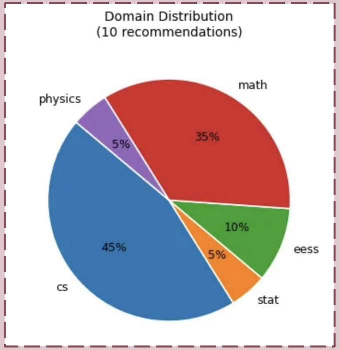
Category Distribution across all models



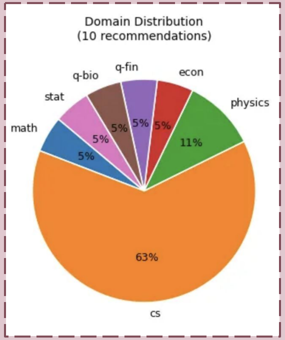
SVD (base)



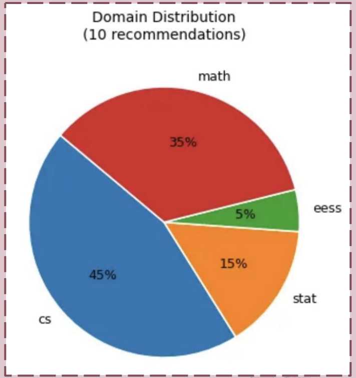
TF-IDF



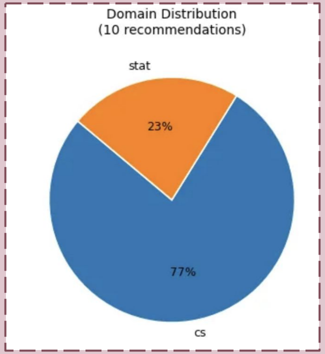
NeuMF



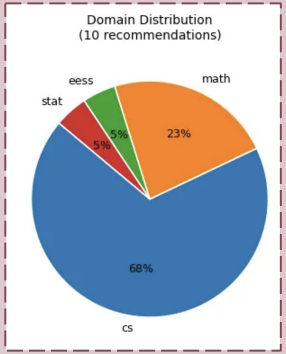
Monte-PPR



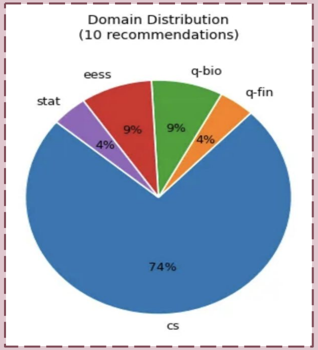
TF-IDF + Monte PPR



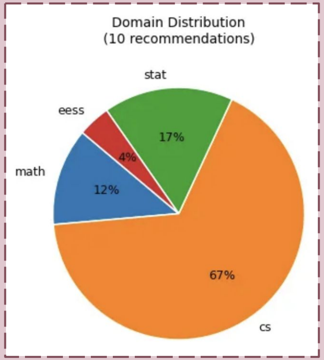
PageRank



GMF



Bandits

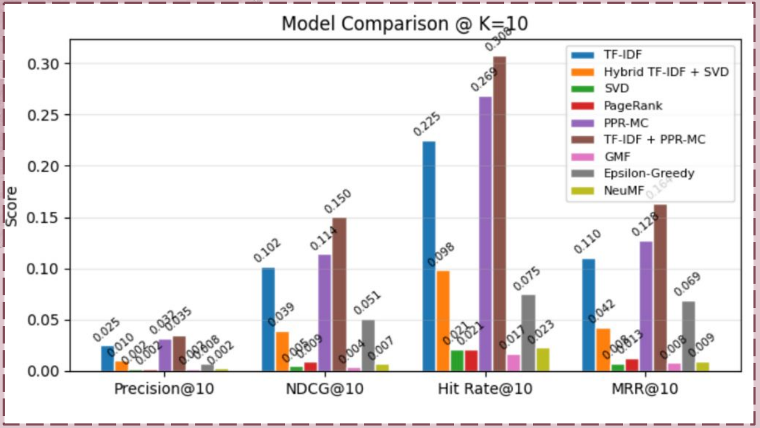


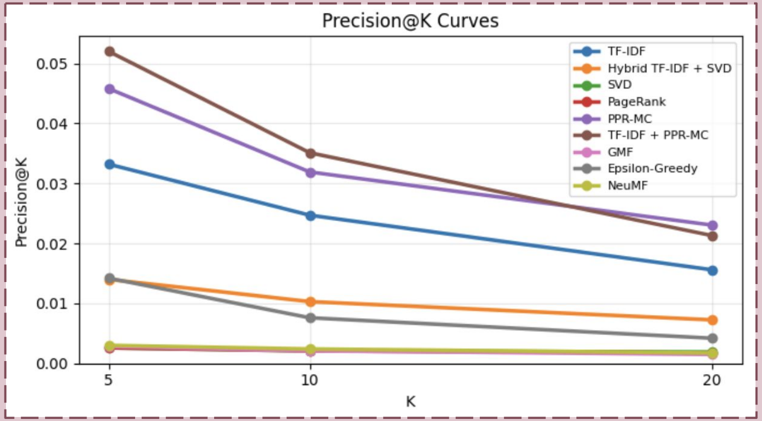
TF-IDF + SVD

Evaluation across all models @ 10

	Prec @10	Recall @ 10	NDCG @10	Hit Rate @10	MRR@10
Baseline (SVD)	0.0021	0.0068	0.0048	0.021	0.0076
TF-IDF	0.0247	0.1486	0.1021	0.225	0.1103
NeuMF	0.0024	0.0093	0.0069	0.023	0.0094
PageRank	0.0021	0.0102	0.0089	0.021	0.0129
Monte PPR	0.0319	0.1632	0.1141	0.269	0.1277
GMF	0.0021	0.0042	0.0038	0.017	0.008
Bandits	0.0076	0.0488	0.0511	0.075	0.0692
Hybrid (tf+svd)	0.0103	0.0592	0.0393	0.098	0.0423
Hybrid (tf + mppr)	0.0351	0.2088	0.1501	0.308	0.1637

Results





- The **best** performing model overall was **TF-IDF + Monte Carlo PPR**, achieving the highest Precision@10 (0.0351), Recall@10 (0.2088), and NDCG@10 (0.1501)
- Monte Carlo PPR alone also performed strongly, showing that graph-based random walk methods are highly effective for citation recommendation
- TF-IDF remained a strong standalone baseline, outperforming most collaborative filtering & neural models
- (SVD, GMF, NeuMF) performed poorly due to sparsity and cold-start limitations
- PageRank & Bandit had moderate performances
- Hybrid models consistently outperformed single-signal models, confirming that combining content + graph signals improves recommendation quality
- Overall, results show that content + graph hybridization is key for strong academic recommendation performance

Conclusion: Key Challenges

Data Creation Issues

- **Metadata format mismatch:** Initial arXiv category extraction failed because the default API format stored categories differently than expected; switching metadata formats resolved filtering
- **Citation data integration issues:** Citation matching initially produced zero valid links due to ID type conversion and stale checkpoint data; fixed through strict string handling and checkpoint validation.

Scalability & evaluation performance

- Baseline recommendation evaluation was too slow at dataset scale and required batching and top-K optimization to reduce runtime
- Multi-hour API collection processes caused session interruptions; atomic checkpointing was implemented for recovery and reproducibility

Cold-start and sparse citation signals

- Low-connectivity papers introduced noise for collaborative models & were filtered to improve recommendation quality (26.48% never cited)
- Due to this, we lost a significant number of papers that could have been valuable information for a user (315,627 papers, 49,041 citations lost)

Conclusion: Key takeaways

Different Algorithms

- **TF-IDF** captured semantic similarity from text
- **SVD, GMF & NeuMF** learned hidden citation relationships
- **PageRank / PPR** leveraged citation graph structure
- **Bandits** used an exploration-exploitation strategy
- All these signals are valuable information in recommendations
- **Cross-domain** search is made possible by these algorithms, as seen in the demo!

Future works

- Including **stronger hybrid** models, neural network methods, etc
- Including the **entire** Arxiv dataset, with the papers w/ degree <2 citations
- Further UI changes, category only recommendations, ability for user to combine models themselves and more

Data Prep

- Recommendation quality depended heavily on clean metadata & citation alignment
- Data validation & checkpointing were necessary for reliable large-scale collection
- **Scalability**: Large academic datasets require efficient preprocessing, sparse representations & scalable evaluation methods

Hybrid Recommender vs Algorithms

- Combining multiple recommendation approaches produced the best result!
- Based on our results, content + graph signals provides the best recommendations
- Typical collaborative algorithms alone struggle against very sparse data, like this particular dataset

Links & Citations

- Github: https://github.com/Talina06/research-paper-recommender-system
- Dataset: https://www.kaggle.com/datasets/Cornell-University/arxiv
- Report: https://docs.google.com/document/d/1gXGpgs15aFqr02k5r2_EwXgNd206JjZ0znkb9CMPD_8/edit?usp=sharing
- This slide deck: https://docs.google.com/presentation/d/1XimiEqJu4G9o4WJ0ZQrzZD4QFo28dsPW60RW06hXYo4/edit?usp=sharing
- *ChatGPT, GPT-5.5 version, OpenAI chatgpt.com*
  - Used for UI, setting up some graphs, fixing model issues

Thank you
Any questions?